Maxmilian Duarte

Python Backend Engineer

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PROFESSIONAL SUMMARY

Python backend engineer with 4 years of production experience building REST and SOAP integrations, browser automation that runs unattended, and data pipelines on advanced SQL. I own systems end to end, from design and tests to Docker deployment on Linux.

SKILLS

Backend: Python, Flask, FastAPI, REST and SOAP APIs, asyncio / aiohttp, OAuth2 / PKCE, Server-Sent Events
Data: MySQL, SQLite, advanced SQL, stored procedures, ETL, Power BI, Adobe Campaign
Infra: Linux (Ubuntu Server, Samba), Docker / Compose, nginx, Proxmox / LXC, GitHub Actions, Uptime Kuma, scheduled backups
Quality: pytest, Playwright end-to-end tests, type hints, production support (L3), log analysis
Also: TypeScript / Next.js, Vue 3, Kotlin / Jetpack Compose, Selenium, FFmpeg, Jira API

WORK EXPERIENCE

Independent Developer — Jul 2026 – Present
Own products and open source — Remote

- Building MaxTools Radio and cnabkit, an open-source Python library for Brazilian bank files (CNAB 240/400, OFX), on self-managed infrastructure: Proxmox, Docker, nginx, Uptime Kuma, and weekly snapshot backups.

Software Engineer (Python, Backend and Data) — Sep 2024 – May 2026
Netbiis | Data Driven Technologies — Remote

- Built an internal Flask and SQL app of operator tools for 32 daily campaign jobs, cutting each from 20 to 12 min (–40%).
- Built a FastAPI validator that checks Jira briefs against Adobe Campaign workflows over SOAP, automating QA on 10 to 15 daily jobs.
- Wrote SQL, stored procedures, JavaScript, and workflow automations in Adobe Campaign for high-volume segmentation.
- Provided L3 support for production automation pipelines: log-based root-cause analysis, fixes, and follow-up.

Backend Developer, Independent Consultant — Feb 2023 – Aug 2024
HG7 Group (Insurance Brokerages and a Clinic) — Hybrid

- Built an OFX bank-statement import and reconciliation system (Python, Flask, MySQL) in three iterations, feeding the monthly income statement, plus commission reconciliation against the brokerage ERP.
- Built async ETL (aiohttp) from RD Station CRM and Exact Spotter into MySQL for Power BI, with retry on HTTP 429.
- Built a bot that reads answered calls from the 3CX call panel and creates RD Station deals assigned to the right salesperson.

Python Developer (Internship) — Aug 2022 – Feb 2024
Plug Benefícios — São Paulo

- Built and maintained 15 production bots (Python, Selenium) entering hundreds of beneficiary changes a month into 4 health-insurer portals from a MySQL ticket queue, plus invoices, slips, and payslips.
- For ExportSul, a client brokerage, built bots on the Akad, Sompo, and Tokio Marine portals that register export shipments and download invoices, resuming where a run stopped.
- Packaged shared code (browser setup, audio-based reCAPTCHA solving, FTP upload) into an internal Python wheel, later reused across other client projects.
- Set up the company's Ubuntu server, replacing Windows Server: Samba file shares, Windows and macOS printing, bot runtime.

Tech Instructor — Oct 2021 – Feb 2022
Microcamp Tecnologia — São Paulo

- Taught programming logic, algorithms, and Python to 50+ students.

PROJECTS

Case studies at maxmilian02.github.io

MaxTools Radio: Python bot streaming 48 kHz PCM through a real-time DSP chain, Flask API with SSE and OAuth2 / PKCE sign-in, web app, and Kotlin / Jetpack Compose Android app. 1,268 Python tests (271 end-to-end in Playwright) and 472 Android tests.

MaxTools: FastAPI plugin platform where a registry auto-discovers tool modules and a Next.js dashboard renders from their manifest. Background jobs with progress and cancellation, API-key auth, TTL cleanup, and frame-accurate video trimming over HTTP Range.

LANGUAGES

Portuguese (native), English (advanced: reading, writing, technical documentation)

EDUCATION

B.Sc. Computer Engineering, Impacta Tecnologia (expected 2027)